Dataset: Kaggle – E-Commerce Dataset
Direct link: https://www.kaggle.com/datasets/carrie1/ecommerce-data

Rows: ~540,000
Tables: 1 table (easy for beginners)
Format: CSV

Columns in dataset

Column	Description
InvoiceNo	Order number
StockCode	Product code
Description	Product name
Quantity	Quantity purchased
InvoiceDate	Date of purchase
UnitPrice	Price per unit
CustomerID	Customer ID
Country	Customer country

Step 1: Create table in MySQL

```
CREATE DATABASE ecommerce_db;

USE ecommerce_db;


CREATE TABLE ecommerce (

InvoiceNo VARCHAR(20),

StockCode VARCHAR(20),

Description TEXT,

Quantity INT,
```

```
InvoiceDate DATETIME,
UnitPrice FLOAT,
CustomerID INT,
Country VARCHAR(50)
);
```

Import CSV using MySQL Workbench.

Step 2: SQL queries

Basic Analysis

1. Total number of transactions

```
SELECT COUNT(*) FROM ecommerce;
```

---

2. Total revenue

```
SELECT ROUND(SUM(Quantity * UnitPrice),2) AS total_revenue
FROM ecommerce;
```

---

3. Total customers

```
SELECT COUNT(DISTINCT CustomerID)
FROM ecommerce;
```

---

4. Total products

```
SELECT COUNT(DISTINCT StockCode)
FROM ecommerce;
```

---

5. Average order value

```
SELECT ROUND(AVG(Quantity * UnitPrice),2)
FROM ecommerce;
```

Intermediate Analysis

6. Revenue by country

```
SELECT Country,
ROUND(SUM(Quantity * UnitPrice),2) AS revenue
FROM ecommerce
GROUP BY Country
ORDER BY revenue DESC;
```

---

7. Top 10 customers by revenue

```
SELECT CustomerID,
ROUND(SUM(Quantity * UnitPrice),2) AS revenue
FROM ecommerce
GROUP BY CustomerID
ORDER BY revenue DESC
LIMIT 10;
```

---

8. Top selling products

```
SELECT Description,
SUM(Quantity) AS total_sold
FROM ecommerce
GROUP BY Description
ORDER BY total_sold DESC
LIMIT 10;
```

---

9. Monthly revenue

```
SELECT DATE_FORMAT(InvoiceDate,'%Y-%m') AS month,
ROUND(SUM(Quantity * UnitPrice),2) AS revenue
FROM ecommerce
GROUP BY month
ORDER BY month;
```

---

10. Orders per country

```
SELECT Country,
COUNT(DISTINCT InvoiceNo)
FROM ecommerce
GROUP BY Country;
```

Advanced SQL Analysis

11. Repeat customers

```
SELECT CustomerID
FROM ecommerce
GROUP BY CustomerID
HAVING COUNT(DISTINCT InvoiceNo) > 1;
```

---

12. Top revenue generating countries

```
SELECT Country,
SUM(Quantity * UnitPrice) AS revenue
FROM ecommerce
GROUP BY Country
```

```
ORDER BY revenue DESC
LIMIT 5;
```

---

13. Highest value transaction

```
SELECT InvoiceNo,
SUM(Quantity * UnitPrice) AS total
FROM ecommerce
GROUP BY InvoiceNo
ORDER BY total DESC
LIMIT 1;
```

---

14. Average quantity per order

```
SELECT AVG(total_quantity)
FROM (
SELECT InvoiceNo,
SUM(Quantity) AS total_quantity
FROM ecommerce
GROUP BY InvoiceNo
) t;
```

---

15. Customers per country

```
SELECT Country,
COUNT(DISTINCT CustomerID)
FROM ecommerce
GROUP BY Country;
```

---

16. Most expensive product

```
SELECT Description,
```

```
MAX(UnitPrice)
FROM ecommerce
GROUP BY Description
ORDER BY MAX(UnitPrice) DESC
LIMIT 1;
```

---

17. Revenue per customer

```
SELECT CustomerID,
SUM(Quantity * UnitPrice) AS revenue
FROM ecommerce
GROUP BY CustomerID;
```

---

18. Total orders per customer

```
SELECT CustomerID,
COUNT(DISTINCT InvoiceNo)
FROM ecommerce
GROUP BY CustomerID;
```

---

19. Best selling country by quantity

```
SELECT Country,
SUM(Quantity)
FROM ecommerce
GROUP BY Country
ORDER BY SUM(Quantity) DESC;
```

---

20. Customers who spent above average

```
SELECT CustomerID,
SUM(Quantity * UnitPrice) AS total_spent
FROM ecommerce
GROUP BY CustomerID
HAVING total_spent >
(
SELECT AVG(total)
FROM (
SELECT SUM(Quantity * UnitPrice) AS total
FROM ecommerce
GROUP BY CustomerID
) t
);
```